

Inter-Animal Contact Geometry from Segmentation Masks

Do mask-derived contact features beat keypoint-hull geometry?

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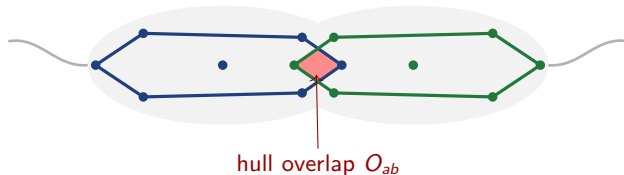
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The question in one line

A **social behaviour** is defined by the **geometry of how two bodies meet** — how close, how much they overlap, where they touch.

- Chase, huddle, sniffing — all are read from *inter-animal geometry*.
- Today the field measures that geometry from **keypoint hulls**.
- **We ask:** can a *segmentation mask* measure it better, exactly where hulls are weakest — *during contact*?

Principle 1 — Hull contact geometry (the field: SimBA, MARS)

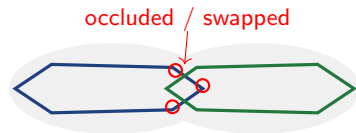


Convex hull $H_i = \text{conv}(K_i)$ over each mouse's **12 keypoints** (dots). Contact = polygon overlap

$$O_{ab} = \frac{|H_a \cap H_b|}{|H_a \cup H_b|} \quad (\text{red}). \text{ Cheap, but built from } \textit{sparse points}.$$

Why hulls are weakest *exactly during contact*

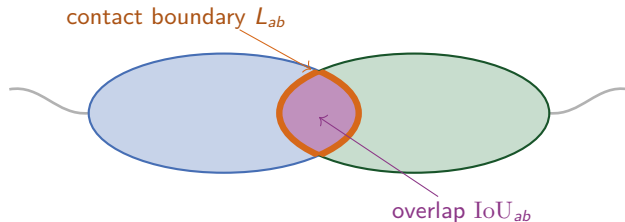
- In close contact the **keypoints occlude** one another.
- On three **near-identical mice** the tracker **swaps / hallucinates** points.
- A polygon over a few noisy points **smooths over the true contact boundary**.



The crux

The representation is **crudest at the very moment** the behaviour is defined.

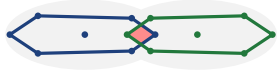
Principle 2 — Mask contact geometry (our proposal, via SAM 2)



Full silhouettes M_a, M_b (SAM 2, prompted by the same keypoints). $\text{IoU}_{ab} = \frac{|M_a \cap M_b|}{|M_a \cup M_b|}$ (plum) and **contact-boundary length** $L_{ab} = |\partial M_a \cap (M_b \oplus B_r)|$ (orange): *how much of the two silhouettes physically touch.*

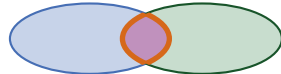
The bet: same contact, two measurements

Hull — sparse points



under-reads the contact

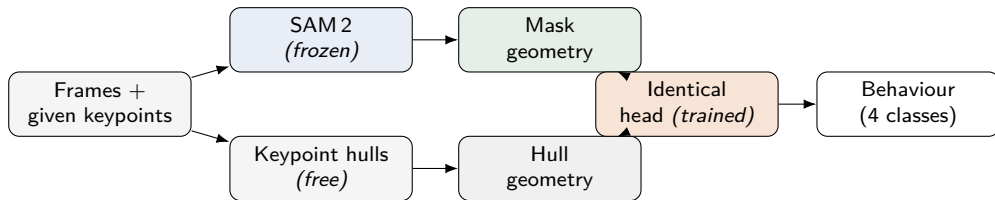
Mask — full silhouette



sees the real overlap + boundary

Both start from the **same keypoints** $\Rightarrow$ the comparison isolates the *representation*, not the pose. **Hypothesis:** the gap is largest on the deepest-contact behaviours.

How it runs: everything heavy is frozen & cached



SAM 2 & VideoMAE are **inference-only**; features cached once. The **only** trained part is a small head of **identical capacity** in every condition — so a gain reflects *information*, not *parameters*.

The data: MABe22 mouse-triplet

- **1,830** labelled 1-min sequences @ 30 Hz (224×224 top-view).
- **3 near-identical mice, 12 given keypoints** each.
- **Dense framewise labels** for four behaviours.
- Top-view lab mice = easy *imaging*; the hard part is *identity through contact*.

Eval contract (already locked): framewise, four independent binary tasks; **leakage-safe split by sequence**; per-behaviour F1 + AP, macro-averaged.

Behaviour	positive
huddles	8.1 %
oral_oral_contact	0.23 %
oral_genital_contact	0.23 %
chases	0.05 %

Heavy imbalance $\Rightarrow$ report **AP** as well as F1.

The experiment (pre-registered)

- a **Hull geometry** alone — the baseline.
 - b **Mask geometry** alone vs (a) — **the key head-to-head**.
 - c **Fusion**: mask geometry on top of frozen VideoMAE — does it add over what VideoMAE already sees?
- Identical head capacity; ≥ 5 seeds; mean $\pm$ 95 % CI; per-behaviour scores.
 - **Success fixed up front**: mask beats hull with a CI excluding zero *and* a pre-set minimum effect, *or* a clear win on the deepest-contact behaviours.
 - Scrambled-mask control rules out a capacity artefact.

Interpretable **either direction** — a clean negative is still a result.

The one make-or-break risk (settled in week 1)

*Can SAM2 keep three near-identical mice separate & identity-stable **through the contact frames?***

PASS

identities hold → proceed;
contact geometry is the headline.

PARTIAL

masks merge only in deepest
contact → use the *merge event itself* as a coarse high-contact feature.

FAIL

frequent swaps → pivot headline to **background-robustness**, report this as a negative result.

A **week-1 masking pilot** on huddle/chase clips fires this decision. The dissertation keeps a clean spine either way.

Eight-week plan

- Wk 1** Acquire + lock eval contract (**done**); **masking pilot** + **decision**.
- Wk 2** Scoped SAM 2 mask cache (Kaggle); begin geometry features.
- Wk 3** Finish contact-geometry + hull extractor; head overfit sanity.
- Wk 4** **Mask vs hull head-to-head** — first real result.
- Wk 5** Per-behaviour analysis; multi-seed CIs; go/no-go on MammAlps.
- Wk 6** Fusion / VideoMAE-redundancy test; lock primary.
- Wk 7** Background-shift robustness (insurance); consolidate.
- Wk 8** Write-up; figures; package pipeline + code.

All risk front-loaded into week 1; the headline result by week 4.

Contribution

To our knowledge, the **first use of inter-animal segmentation-mask contact geometry** (mask IoU, contact-boundary length) as an explicit feature for social-behaviour recognition — in a clean head-to-head against the keypoint-hull geometry the field already uses.

- Honest framing: shape-from-silhouette is *not* new; the **mask-contact measurement** and the **controlled comparison** are.
- Every heavy model frozen; reproducible from cache; multi-seed CIs.
- Built-in insurance: background-robustness if the masking pilot fails.

Thank you — questions welcome.